



Ben Youcef Benkheda University

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Department of Computer Science,
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MI. L 1. Sections A & B

Module: Algebra 1

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THANKS IN ADVANCE

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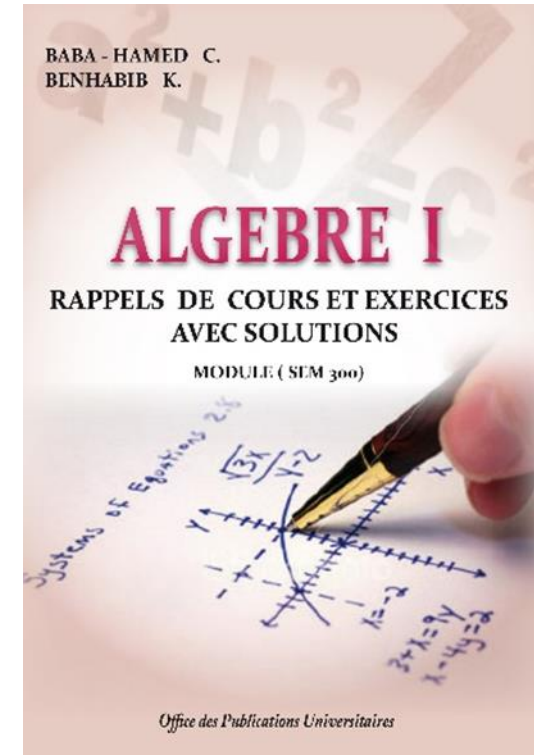
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Algebra 1 : the seven chapters

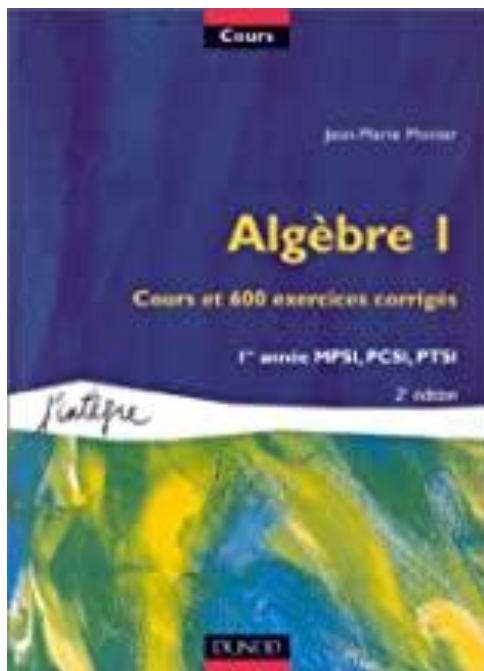
- 1) Introduction to Mathematical Logic
- 2) Notion of Set
- 3) Applications
- 4) Binary Relations
- 5) Algebraic Structures: Groups, Rings and Fields
- 6) The field of Complex numbers : $\mathbb{C}$
- 7) Polynomials

BIBLIOGRAPHY (1/3)

- › C,BABA HAMED
- › K,BENHABIB
- › **ALGÈBRE I**
- › RAPPELS DE COURS ET EXERCICES AVEC SOLUTIONS,
- › O.P.U.



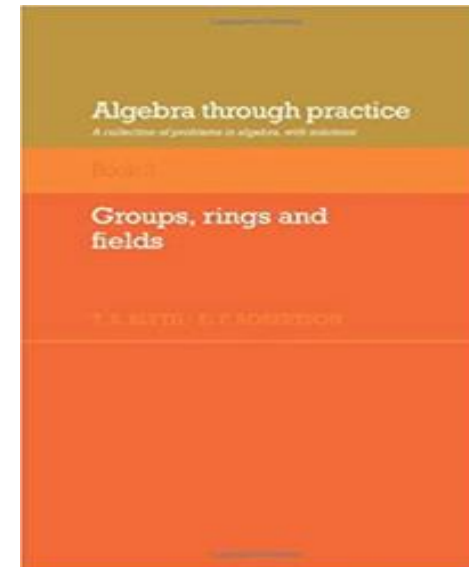
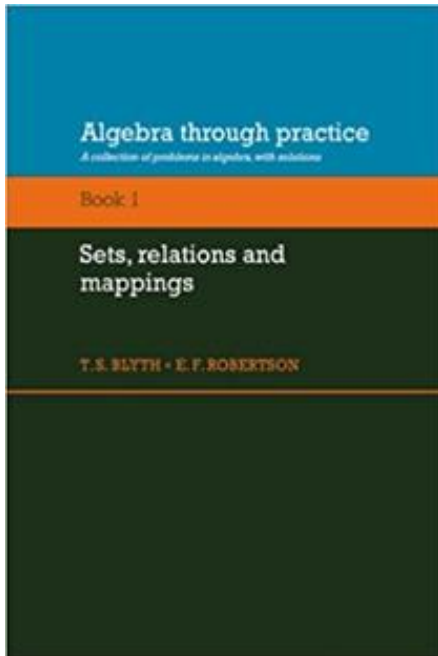
BIBLIOGRAPHY (2/3)



Algèbre, tome 1 : Cours et 600 exercices corrigés, 1re année MPSI, PCSI, PTSI
de [Jean-Marie Monier](#)

BIBLIOGRAPHY (3/3)

- › **T. S. Blyth :**
- › **Algebra through Practice: Volume 1,2,3: A Collection of Problems in Algebra with Solutions**



SITES WEB

- › <http://exo7.emath.fr/un.html>



- › <http://www.bibmath.net/ressources/index.php?action=affiche&quoi=bde/index>

Bibm@th.net

Preamble

INTRODUCTION

AL KHAWARIZMI



 Vedic Math School
MANGAL MATHS KATG...

**Interesting facts about
Al-Khwarizmi**

1. He is the father/ founder of algebra.
2. He worked in Baghdad at the House of Wisdom as a scholar.
3. He found out the circumference of the Earth.
4. He wrote the trigonometric functions table of cosine and sines.

The Compendious Book on Calculation by Completion and Balancing

(Arabic: الكتاب المختصر في حساب الجبر والمقابلة, *al-Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wal-Muqābalaḥ*;

Latin: *Liber Algebræ et Almucabola*), also known as **al-Jabr**

(Arabic: الجبر) is an Arabic mathematical treatise on algebra written in Baghdad around 820 CE by the Persian polymath Muhammad ibn Mūsā al-Khwārizmī.

It was a landmark work in the history of mathematics, establishing algebra as an independent discipline.



The *Ḥisab al-jabr w'almuqabala*

The treatise about Algebra "*Ḥisab al-jabr w'al-muqabala*" was the most famous and important of all of Al-Khwarizmi's works. It is the title of this text that gives us the word "algebra" and it is the first book to be written about Algebra.

Chapter 1

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Introduction to mathematical logic

Begin at the beginning,' the King said gravely, 'and go on till you come to the end: then stop'.



Charles Lutwidge **Dodgson**, aka Lewis **Carroll**

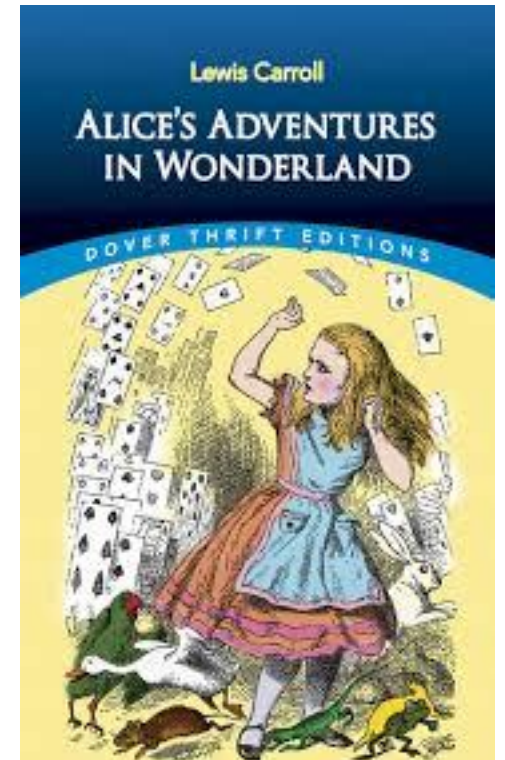
British logician and writer

(Daresbury, Cheshire, 1832-Guildford 1898).

The White Rabbit put on **his** spectacles. '**Where shall I begin, please your Majesty?**' he asked.

'Begin at the beginning,' the King said gravely,

'and go on till you come to the end:
then stop'.



Goals of this course

In this first part, the aim is to formulate more rigorous definitions of mathematical concepts already encountered in high school.

Objectives of this module.

Acquire new terminology,

Facilitation with different reasoning methods.

Mastery of the use of mathematical symbols.

Learning the correct writing of mathematical language.

The adopted approach.

- › In order to make the new terminology more accessible and to familiarize ourselves with the new concepts, we will draw, in particular for the examples, on the previous knowledge acquired in high school.

Chapter 1 Schedule (1/2):

- › Notions of Statement (or Proposition) and Truth Values.
- › Unary, binary logical connectives.
- › Truth tables.
- › “Electrical” analogies.
- › De Morgan’s laws.

Chapter 1 Schedule (2/2):

- › Example of Tautology, Antilogy.
- › Notions of set, element and membership.
- › Subset, Power Set.
- › Cartesian product.
- › Venn diagram.
- › Predicates, quantifiers.

The definition

- › In Mathematics, knowing the definition of a new object is fundamental.
- › Knowing examples of this new object makes its understanding easier.

Example of a mathematical definition

› Definition:

A natural number p is said to be prime if it is different from 1, and if its only divisors are 1 and p itself.

› Examples:

2, 3, 5, ..., 19, ..., 691 are prime numbers.

› Remarks:

4, 27, 91, 135, $2^{67}-1$ are not prime numbers, they are said to be ***composite numbers***.

The Theorem

- › A theorem is the statement of a proposition the truth of which we propose to establish using a demonstration (or a proof).
- › The theorem consists of a hypothesis (supposition) and a conclusion (result).
- › *The lemma* is a "small theorem" used ad hoc to prove an important theorem.

The demonstration (or proof)

- › The demonstration is a sequence of logical deductions developed from previously formulated definitions and axioms.
- › The demonstration serves to prove a result stated in a theorem.

The Statement

› A statement (or *proposition*, or *atomic statement*) is any phrase (or assertion, or sentence) that is either

true (symbolized by T or 1)

or

false (symbolized by F or 0)).

but not both simultaneously.

› T (or 1), F (or 0) are called truth values.

Example

- › Let P, Q and R be the following three statements:
- › P: "19 is a prime number".
- › Q: "1437 is even".
- › R: "Is this car fast?".
- › We observe that:
- › *P is a true statement.*
- › *Q is a false statement.*
- › *R is not a statement.*

Compound statement

- › Based on several propositions (or atomic statement),it is possible to create new ones by connecting (or assembling, or compound) them using “logical connectives”.

The most used logical connectives

- › In mathematics, the most used logical connectives are:
- › **negation,**
- › **conjunction,**
- › **disjunction,**
- › **implication**
- and
- › **equivalence.**
- › There are others like NAND, NOR and XOR.

The logical connectives and truth tables

- › These connectives can be defined by words of a language (English, French or Arabic, or...etc) or by what we call **truth tables**.

The negation

Let A be a proposition. The negation of A , symbolized by " $\bar{A}$ " (or "not A ", or "no A "), is the proposition which is true if A is false and false if A is true; which results in the following truth table,

Truth Table of the negation	
A	No A ($\bar{A}$)
V	F
F	V

Examples of negations

› **A: « Algiers is the capital of Algeria ».**

› $\bar{A}$: « **Algiers is not the capital of Algeria** ».

› **B: « 19 is even ».**

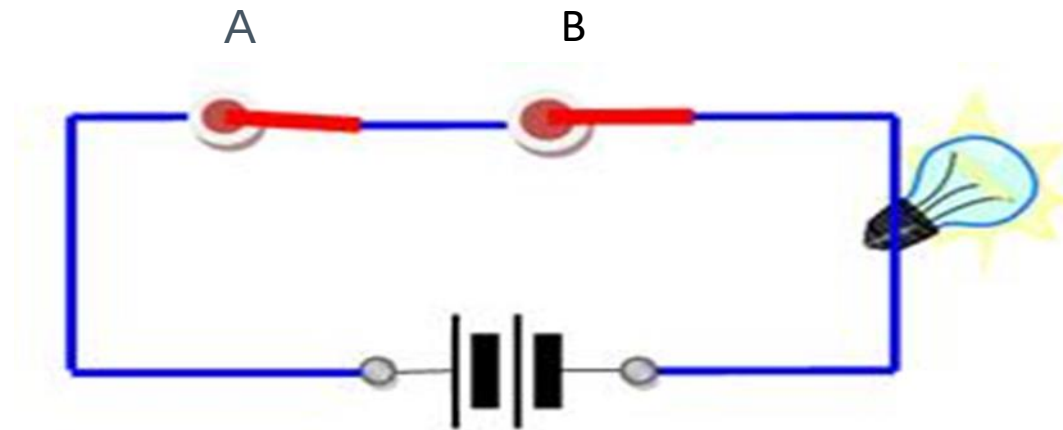
› $\bar{B}$: « **19 is not even** ».

The conjunction (symbolized by $\wedge$ or *and*) defined by the following truth table:

A	B	A $\wedge$ B
V	V	V
V	F	F
F	V	F
F	F	F

Electric circuit analogous to the conjunction

Switches A and B are the electrical analogue of the statements



Example :

A : « 6 is odd »

B: « 16 is a square number ».

A is false

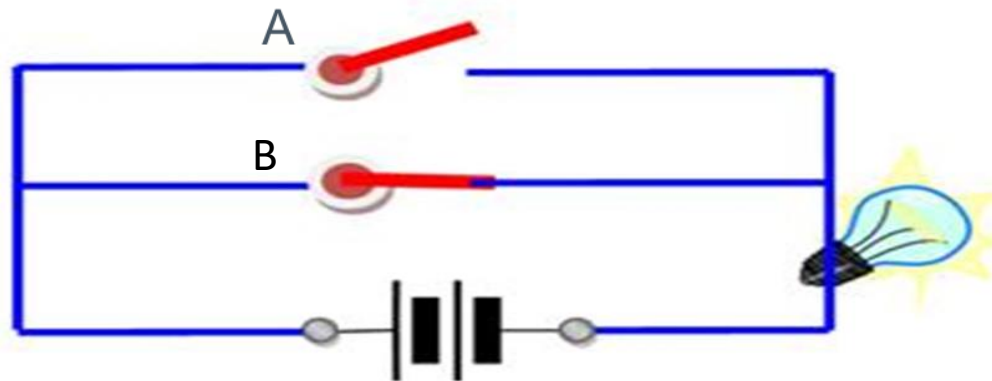
B is true

but $(A \wedge B)$ is false.

Truth table of the disjunction (with the previous notations)

A	B	$A \vee B$
V	V	V
V	F	V
F	V	V
F	F	F

Electric circuit analogous to the disjunction



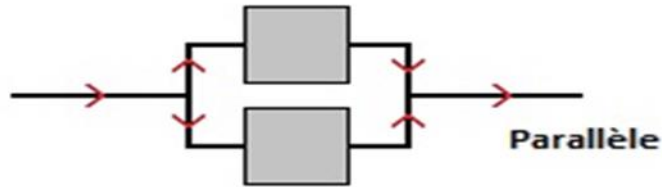
Comparing disjunction and conjunction

Series connection



Conjunction

Parallel connection



Disjunction

The implication : $\Rightarrow$

Truth Table of the implication

A	B	$A \Rightarrow B$
V	V	V
V	F	F
F	V	V
F	F	V

Others wordings of the implication (1/3)

› In the following implication:

$$\mathbf{A} \Rightarrow \mathbf{B}$$

- › $\mathbf{A}$ is called **sufficient condition** or *hypothesis*.
- › $\mathbf{B}$ is called **necessary condition** or *conclusion*.

or

- › To have $\mathbf{B}$, it is sufficient that $\mathbf{A}$.
- › To have $\mathbf{A}$, it is necessary that $\mathbf{B}$.

Others wordings of the implication (2/3)

› Example :

› *Let n be a natural number.*

Consider the implication:

$$(n \geq 5) \implies (n \geq 2)$$

› Thus :

› $(n \geq 5)$ is a sufficient condition

› And

› $(n \geq 2)$ is a necessary condition.

Others wordings of the implication (3/3)

› Example :

› *Let n be a natural number.*

Consider the implication:

$$(n \geq 5) \implies (n \geq 2)$$

a) For n to be greater than or equal to 2, it is sufficient for n to be greater than or equal to 5.

b) For n to be greater than or equal to 5, it is necessary that n must be greater than or equal to 2.

Others wordings of the implication (3/3)

› Example :

› *Let n be a natural number.*

Consider the implication:

$$(n \geq 5) \implies (n \geq 2)$$

1) if the natural number n is greater than or equal to 5, then it is greater than or equal to 2.

2) If $(n \geq 5)$, then $(n \geq 2)$.

3) $(n \geq 5)$ is a sufficient condition and
 $(n \geq 2)$

is a necessary condition.

The Contrapositive, Converse, and Inverse of an Implication

Definition:

Let P and Q be statements and consider the implication

$$P \Rightarrow Q.$$

The **Contrapositive** of this implication is the formula $\bar{Q} \Rightarrow \bar{P}$.

The **Converse** of this implication is the formula $Q \Rightarrow P$.

The Contrapositive, Converse, and Inverse of an Implication, Example:

The **Converse** of $(n \geq 5) \Rightarrow (n \geq 2)$ is
 $(n \geq 2) \Rightarrow (n \geq 5)$.

The statement $(n \geq 5) \Rightarrow (n \geq 2)$ is true

but

its **Converse**

$$(n \geq 2) \Rightarrow (n \geq 5)$$

is false.

The **Contrapositive** of $(n \geq 5) \Rightarrow (n \geq 2)$ is :

$$(n < 2) \Rightarrow (n < 5) ,$$

which is still true.

Logical equivalence ($\Leftrightarrow$)

- › Two statements **A** and **B** are said to be equivalent if they have the same truth values.
- › Notation:

$$\ll \mathbf{A} \Leftrightarrow \mathbf{B} \gg$$

Or also in some textbooks:

$$\ll \mathbf{A} \equiv \mathbf{B} \gg$$

The equivalence ($\Leftrightarrow$, iff)

› The logical equivalence , denoted by

« $\Leftrightarrow$ »,
is defined by the following truth table :

A	B	$A \Leftrightarrow B$
1	1	1
1	0	0
0	1	0
0	0	1

And can be read : if and only iff.

Exercise: show that :

$(A \Leftrightarrow B)$

is the same as :

$((A \Rightarrow B) \text{ and } (B \Rightarrow A))$

« Definition has the same logical strength as an equivalence »

- › In Mathematics, a definition "has the status" of an equivalence. When stating a definition, one is not obliged to write

"If and Only if"

which is therefore always implied in a definition.

Example:

A definition has the same logical force as an equivalence"

Consider the following definition:

- › "***A triangle is isosceles if it has two equal sides***"
- › This also means that if a triangle has two equal sides, then it is isosceles.
- › In middle school, we prove the theorem:
- › "A triangle is isosceles if and only if it has two equal sides."

De Morgan's Laws

(1806-1871)



› A and B being two statements, show that:

$$\text{a) } \overline{(A \vee B)} \Leftrightarrow (\bar{A} \wedge \bar{B})$$

$$\text{b) } \overline{(A \wedge B)} \Leftrightarrow (\bar{A} \vee \bar{B})$$

Distributivity Laws

› A, B and C being three statements, show that:

$$1) (A \vee (B \wedge C)) \Leftrightarrow ((A \vee B) \wedge (A \vee C))$$

$$2) (A \wedge (B \vee C)) \Leftrightarrow ((A \wedge B) \vee (A \wedge C))$$

The main types of reasoning (1/2)

A , B and C being three statements, show that:

1. Transitivity :

From $(A \Rightarrow B)$ true and $(B \Rightarrow C)$ true, one can deduce that
 $(A \Rightarrow C)$ true.

2. Syllogism :

From A true and $(A \Rightarrow B)$ true, one can deduce that B is true.

The main types of reasoning (2/2)

› 3. Disjunction of cases

To demonstrate a property, it is sometimes necessary to study case by case.

This reasoning is called "disjunction of cases".

To demonstrate $P \Rightarrow Q$, we decompose into n subcases and demonstrate $P_1 \Rightarrow Q, P_2 \Rightarrow Q, \dots, P_n \Rightarrow Q$.

4. Use of the contrapositive:

To prove $P \Rightarrow Q$, sometimes it is more easier to prove :

$$\bar{Q} \Rightarrow \bar{P}$$

The main types of reasoning

Proof by contradiction

- › **Proof by contradiction** is a method of proving mathematical statements based on the law of the excluded middle in logic. *The law of excluded middle states that there are only two possibilities, either a statement is true or its negation is true.*
- › In **proof by contradiction** we assume the negation of a statement and proceed to show that the assumption leads us to a contradiction or an absurdity, thus concluding that our statement is, in fact, true. It is also known as **indirect proof** or the method of *reductio ad absurdum*.

Explanation of proof by contradiction

› To prove that $P \Rightarrow Q$ is true and as we know that

$P \Rightarrow Q$ is equivalent to $\bar{P} \text{ or } Q$,

we will assume that the negation of $P \Rightarrow Q$, which is

$P \text{ and } \bar{Q}$

is true.

We end up with a contradiction therefore $P \text{ and } \bar{Q}$

is false, so its negation $\bar{P} \text{ or } Q$ is true,

therefore $P \Rightarrow Q$ is true.

Proof by contradiction : EXAMPLE

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For each real number x , if $0 < x < 1$, then $\frac{1}{x(1-x)} \geq 4$

Proof

We will use a proof by contradiction. So we assume that the proposition is false, or that there exists a real number x such that $0 < x < 1$ and

$$\frac{1}{x(1-x)} < 4. \quad (1)$$

We note that since $0 < x < 1$, we can conclude that $x > 0$ and that $(1-x) > 0$. Hence, $x(1-x) > 0$ and if we multiply both sides of inequality (1) by $x(1-x)$, we obtain

$$1 < 4x(1-x).$$

We can now use algebra to rewrite the last inequality as follows:

$$1 < 4x - 4x^2$$

$$4x^2 - 4x + 1 < 0$$

$$(2x - 1)^2 < 0$$

However, $(2x - 1)$ is a real number and the last inequality says that a real number squared is less than zero. This is a contradiction since the square of any real number must be greater than or equal to zero. Hence, the proposition cannot be false, and we have proved that for each real number x , if

$$0 < x < 1, \text{ then } \frac{1}{x(1-x)} \geq 4.$$

Proof by induction

› A proof by induction of $P(n)$, a mathematical statement involving a **natural number n** , involves these **three** main steps:

- **The Base case:**

Prove directly that $P(n)$ is correct for the initial value of n (for most examples you will see this is zero or one).

- **The induction hypothesis**

Assume for some natural number k that $P(k)$ is correct..

- **The induction step.**

We will now prove directly that $P(k) \Rightarrow P(k + 1)$. That means prove directly that $P(k + 1)$ is correct by using the fact that $P(k)$ is correct.

The combination of these three steps shows that $P(n)$, is true for all values of n .

Example of reasoning by induction (1/2)

- › Prove by induction that $n^3 + 2n$ is divisible by 3 for every non-negative integer n .

Let $P(n)$ be the mathematical statement $n^3 + 2n$ is divisible by 3, i.e.
 $P(n)$: $\ll n^3 + 2n \text{ is divisible by } 3 \gg$.

- › **Base Case:**

When $n = 0$, we have $0^3 + 2 \cdot 0 = 0$. So $P(0)$ is correct.

Induction hypothesis:

Assume that $P(n)$ is correct for some positive integer k . That means $k^3 + 2k$ is divisible by 3 and hence that $k^3 + 2k = 3m$ for some integer m .

Example of reasoning by induction(2/2)

› Induction step:

We will now show that $P(k + 1)$ is correct. So we will take the original formula and replace the n with $k + 1$ to get $(k + 1)^3 + 2(k + 1)$ and we will show that this is divisible by 3.

At some stage in the proof we will need to use the fact that $k^3 + 2k = 3m$,
so when we re-arrange the formula we will try to get $k^3 + 2k$ as part of it.

We get:

$$\begin{aligned}(k + 1)^3 + 2(k + 1) &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \text{ “by expanding the brackets”} \\ &= k^3 + 2k + 3(k^2 + k + 1) \text{ “by re-arranging the formula”} \\ &= 3m + 3(k^2 + k + 1) \text{ “by the induction hypothesis”} \\ &= 3(m + k^2 + k + 1) \text{ since both parts of the formula have a common factor of 3.}\end{aligned}$$

As $m + k^2 + k + 1$ is an integer, we have that $(k + 1)^3 + 2(k + 1)$ is divisible by 3,
so $P(k + 1)$ is correct.

Hence by mathematical induction $P(n)$ is correct for all non-negative integers n .

Exercises with and without solutions

› Exercise 1 (corrigé) : équivalence logique

Let A and B be two statements.

Show that the following two statements are equivalent:

$$(A \Leftrightarrow B)$$

and

$$((A \Rightarrow B) \wedge (B \Rightarrow A))$$

(Sometimes these two compound statements are to be said : logically equivalent and we denote this fact as:

$$(A \Leftrightarrow B) \equiv ((A \Rightarrow B) \wedge (B \Rightarrow A))$$

Solution of the exercise 1

- › The solution is given by the last two columns of the following truth table

A	B	$A \Rightarrow B$	$B \Rightarrow A$	$(A \Rightarrow B) \wedge (B \Rightarrow A)$	$A \Leftrightarrow B$
V	V	V	V	V	V
V	F	F	V	F	F
F	V	V	F	F	F
F	F	V	V	V	V



Exercise 2 :

Let **A** and **B** two statements.

Show that :

$$1) (A \Rightarrow B) \equiv (\text{no}(A) \vee B)$$

(This result may be taken as the definition of the implication.)

$$2) ((\text{no}(A \Rightarrow B)) \equiv ((A) \wedge \text{no}(B))) \quad (\text{negation of the implication})$$

$$3) (A \Rightarrow B) \equiv (\text{no}(B) \Rightarrow \text{no}(A)) \quad (\text{Contraposition})$$

Tautology

› Definition:

A tautology is a statement which is always true whatever the truth value of the statements which constitute it.

Exemples :

1. $\bar{\bar{A}} \equiv A$
2. $A \vee \bar{A}$. (« *Law of excluded middle* »).
3. $\overline{(A \wedge (\bar{A}))}$. (« *Law of no-contradiction* »).

Exercise 3 : A, B and C being three statements

- › 1) Show, using truth tables, that the following propositions are tautologies:
- › a) **$\text{no}(\text{no}(A)) \Leftrightarrow A$** (*double negation*)
- › b) **$A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$** (*distributivity*)
- › c) **$(A \wedge B) \Rightarrow A$**
- › d) **$A \Rightarrow (A \vee B)$**
- › e) Is implication commutative? Associative?

Exercise 4 :

- › Given three statements P, Q and R, prove the following tautologies:
- › $(P \vee (Q \wedge R)) \equiv ((P \vee Q) \wedge (P \vee R))$
- › $(P \wedge (Q \vee R)) \equiv ((P \wedge Q) \vee (P \wedge R))$

Antilogy (or contradiction)

- › Définition:
- › We call antilogy a statement which is always false whatever the truth value of the statements which constitute it.

- › Example :

$$A \wedge \textit{no}(A).$$